- Similar to thermodynamic principles, electrochemical principles also find applications in metallurgical process.
- If E^0 is positive then the ΔG is negative and the reduction is spontaneous and hence a redox reaction is planned in such a way that the e.m.f of the net redox reaction is positive. When a more reactive metal is added to the solution containing the relatively less reactive metal ions, the more reactive metal will go into the solution.
- Generally the metal extracted from its ore contains some impurities such as unreacted oxide ore, other metals, nonmetals etc...Removal of such impurities associated with the isolated crude metal is called refining process.

EVALUATION

92YZB

Choose the correct answer:

1. Bauxite has the composition

   a) Al_2O_3 b) $Al_2O_3.nH_2O$ c) $Fe_2O_3.2H_2O$ d)None of these

2. Roasting of sulphide ore gives the gas (A).(A) is a colourless gas. Aqueous solution of (A) is acidic. The gas (A) is

   a) CO_2 b) SO_3 c) SO_2 d) H_2S

3. Which one of the following reaction represents calcinations?

   a) $2Zn + O_2 \longrightarrow 2ZnO$ b) $2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$

   c) $MgCO_3 \longrightarrow MgO + CO_2$ d)Both (a) and (c)

4. The metal oxide which cannot be reduced to metal by carbon is

   a) PbO b) Al_2O_3 c) ZnO d) FeO

5. Which of the metal is extracted by Hall-Heroult process?

   a) Al b) Ni c) Cu d) Zn

6. Which of the following statements, about the advantage of roasting of sulphide ore before reduction is not true?

   a) ΔG_f^0 of sulphide is greater than those for CS_2 and H_2S.

   b) ΔG_r^0 is negative for roasting of sulphide ore to oxide

   c) Roasting of the sulphide to its oxide is thermodynamically feasible.

   d) Carbon and hydrogen are suitable reducing agents for metal sulphides.

XII U1 Metallurgy - Jerald.indd 20 16-12-2022 18:53:05